

Custom SVM Experiment - Full Code

Complete Python Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.svm import SVC
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import matplotlib.pyplot as plt
import seaborn as sns

iris = load_iris()

features = iris.data
labels = iris.target

X_train, X_test, y_train, y_test = train_test_split(
    features,
    labels,
    test_size=0.25,
    random_state=10,
    stratify=labels
)

scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)

classifier = SVC(kernel="rbf", C=1.0, gamma="scale")
classifier.fit(X_train, y_train)

predictions = classifier.predict(X_test)

accuracy = accuracy_score(y_test, predictions)
matrix = confusion_matrix(y_test, predictions)

print("Support Vector Machine (SVM)")
print("Accuracy :", accuracy)
print(matrix)
print(classification_report(
    y_test,
    predictions,
    target_names=iris.target_names
))

plt.figure(figsize=(6,5))
sns.heatmap(matrix,annot=True,fmt="d",cmap="Blues",
            xticklabels=iris.target_names,
            yticklabels=iris.target_names)
plt.title("SVM Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
```

Expected Output

Accuracy: 1.00

Classification Report: Precision, Recall and F1-score are 1.00 for all classes.

Output Graph

